



horus eye

AI-Based Exam Proctoring & Monitoring System

TED UNIVERSITY
Department of Computer Engineering
CMPE 491 / 492
Senior Design Project
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Detects. Explains. Never decides.

Detects 17 incident behaviours · Explains every alert with rule + evidence · Defers the final verdict to the human proctor

THE PROBLEM

Conventional in-person exams place one proctor in front of 30+ students. Humans cannot sustain reliable attention for short, localised cheating attempts — a glance at a neighbour, a phone briefly produced, a whispered exchange.

Post-exam video review takes longer than the exam itself, and proctors who suspect an incident often cannot produce the timestamped evidence required by academic-integrity boards.

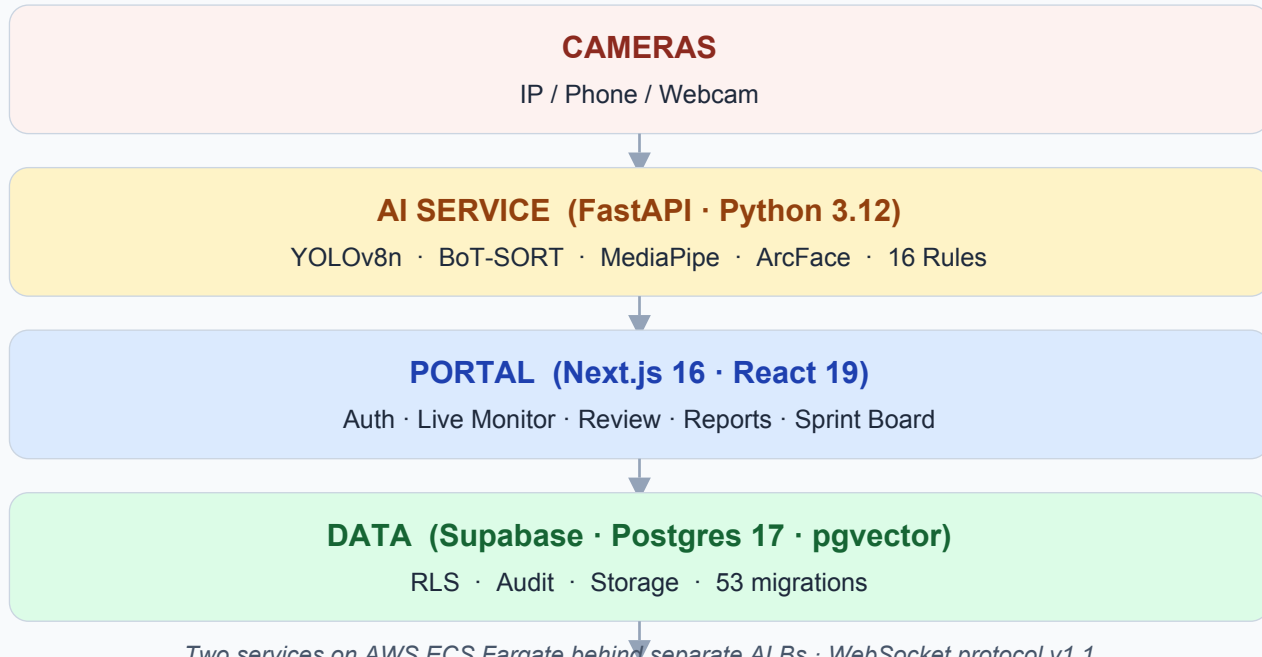
OUR APPROACH

- Real-time AI assistant: not a replacement for the proctor.
- Multi-signal fusion: object detection + tracking + gaze + pose + identity.
- Every alert is explainable: rules and evidence shown to the human.
- Decision authority remains with the proctor: clean / suspicious / violation.
- CPU-only inference — no GPU required; ~\$60/month cloud cost.
- KVKK / GDPR-compliant: minimal retention, consent at sign-in.

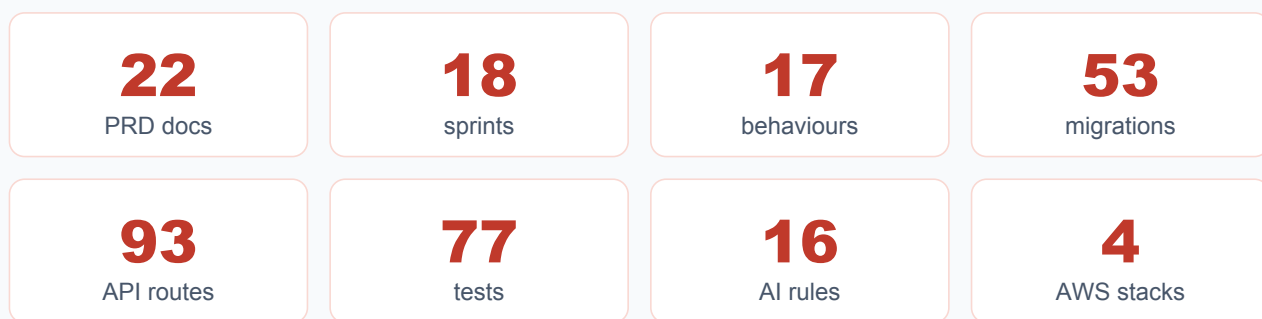
KEY FEATURES

- Multi-step exam wizard with room, camera and student assignment
- Live monitor with per-track incident ring + bbox overlay
- Post-exam review with clean / suspicious / violation modal
- Multi-camera + telephone-as-camera support
- PDF report per exam, session and student
- Sprint platform with PRD-linked backlog (PRD-018)

SYSTEM ARCHITECTURE



BY THE NUMBERS



SPRINT TIMELINE



ENGINEERING DISCIPLINE

- 22 PRDs, versioned interface contracts**
 - Pre-commit hook blocks drift between PRD-000 and consumer PRDs.
- Sprint backlog with dependency enforcement**
 - Blocker not 'done' prevents dependant from advancing (409 error).
- Code-review workflow + audit trail**
 - BacklogReview row required before status='done'.
- CI gates: secrets · env-vars · types · tests · build**
 - GitHub Actions runs Supabase migrations BEFORE Docker build.
- Zero-downtime ECS rolling deploys**
 - minHealthyPercent=100 keeps old container until new healthy.

AI PIPELINE

- Frame**
5 FPS · JPEG
- YOLOv8n**
person · phone · book
- BoT-SORT**
multi-track + reID
- MediaPipe**
FaceMesh · Pose
- ArcFace**
512-D embedding
- Rules × 16**
temporal windows
- Incident**
evidence + WS push

17 BEHAVIOURS DETECTED

- phone in hand ● paper detected ● empty seat ● unauthorized person
- gaze diversion ● head turn ● hand to ear mouth ● hand under desk
- body lean neighbor ● standing up ● object passing ● gaze at lap
- gaze at neighbor ● synchronized behavior ● face covering ● position uncertainty
- whispering (Phase C)
- Severity: ● low ● medium ● high ● critical

APP IN USE



TECHNOLOGY STACK

Next.js 16 · React 19 · TypeScript · Tailwind v4 · shadcn/ui · Supabase · Postgres 17 · pgvector · FastAPI · Python 3.12 · YOLOv8n · BoT-SORT · MediaPipe · InsightFace · OpenCV · AWS CDK · ECS Fargate · ALB · Route 53 · GitHub Actions · Serwist PWA

Social Impact · Ethics · KVKK — the system never makes a verdict on its own

Every detected behaviour is a candidate for the human proctor — we built HorusEye for fair education, not for automated surveillance.

WHO BENEFITS

- Students — fair, transparent grading
- Faculty — relief from attention-fatigue
- Institutions — defensible academic-integrity decisions
- Society — sustained trust in academic degrees

RESPONSIBLE DESIGN

- Human-in-the-loop: proctor decides, AI never issues a verdict
- Every alert is explainable (rule + evidence + timestamp)
- CPU-only at ~\$60/month → affordable for public schools
- Open architecture, no vendor lock-in, no proprietary cameras

PRIVACY & COMPLIANCE

- KVKK Madde 11 · GDPR Article 22 reflected in data model
- Consent captured at sign-in; revocable at any time
- Face embeddings processed only during active session
- Incident evidence purged after 30 days
- Audit logs append-only — tamper-evident

Aligned with UN Sustainable Development Goals · SDG 4 Quality Education · SDG 10 Reduced Inequalities · SDG 16 Strong Institutions

THE TEAM

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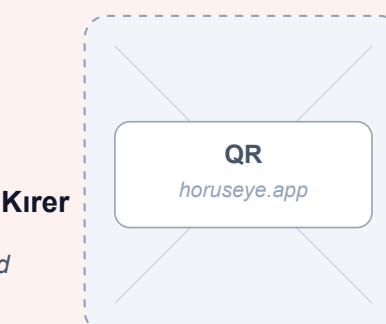
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github.com/kursatozturk/HorusEye
horuseye.app